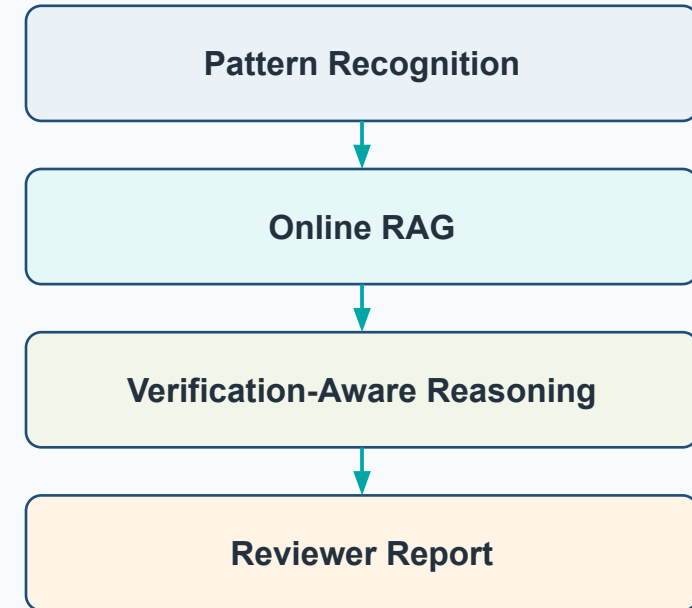




Agentic Clinical Pattern Recognition and Decision Support Platform

Evidence-aware healthcare claim review support - not automated adjudication



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Selected Topic and Interpretation

Clinical Decision Making and Pattern Recognition in Healthcare

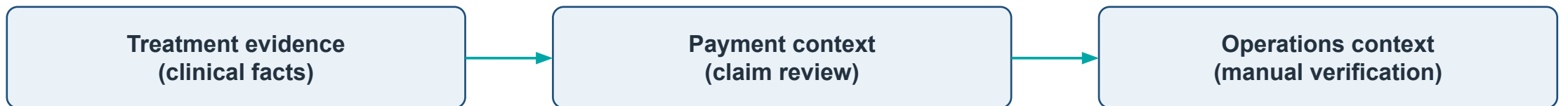
Official topic focus

Chain reasoning, agentic generative AI, classification, prediction, inference, clustering, and anomaly detection for Treatment, Payment, and Operations.

Our scoped interpretation

ClinIQ focuses on decision support for healthcare claim review: pattern recognition, evidence retrieval, verification-aware reasoning, and reviewer recommendations.

Topic mapping



Problem: Evidence Review Is Hard to Scale

Claim reviewers must compare clinical facts, coding, policy, and verification gaps.

Manual burden

Reviewers must inspect diagnosis, procedures, medical necessity, payer policy, benefits, provider status, and authorization records.

Information gaps

Public evidence can support coding and clinical context, but internal payer systems are needed for member benefits, authorization, and provider validation.

Risk of over-automation

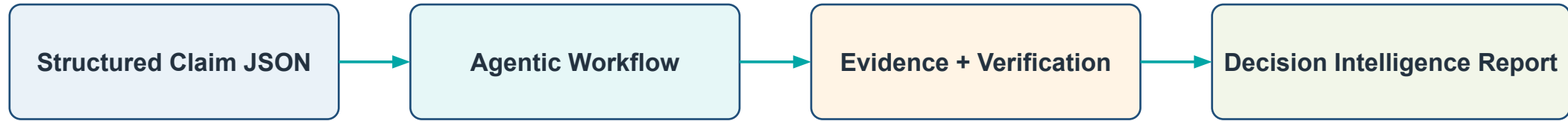
A system that hides uncertainty can produce unsafe or unsupported conclusions. Human review must remain final.

Design response

Evidence first. Recommendations second. Human decision ultimatum.

ClinIQ Product Vision

A decision intelligence assistant for evidence-aware healthcare claim review.



What ClinIQ does

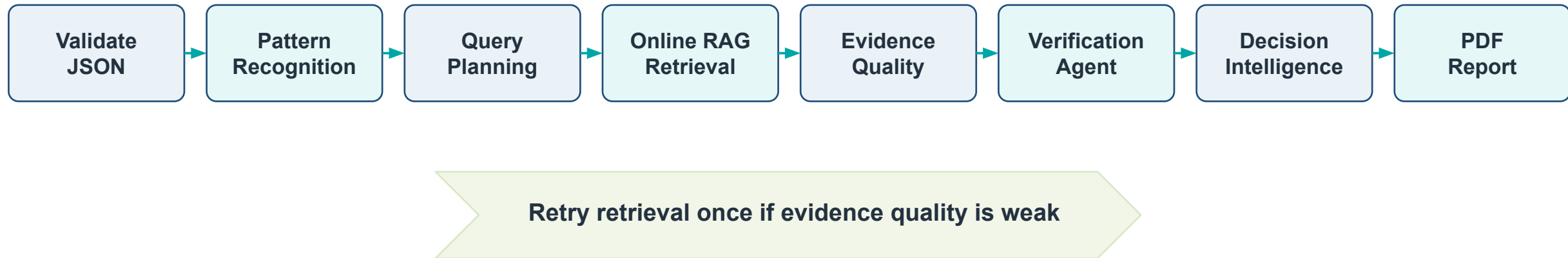
- Detects claim-level clinical and coding patterns
- Retrieves public evidence with online RAG
 - Separates public evidence from internal verification needs
- Produces reviewer-oriented recommendations only

What ClinIQ does not do

- Does not approve, deny, or adjudicate claims
- Does not access private payer/provider systems
- Does not replace authorized reviewers

System Architecture

The prototype demonstrates the full evidence-analysis chain.



Main idea

The project is not a single LLM prompt. It is a chain of specialized stages that inspect, retrieve, verify, reason, and report.

Agentic Workflow

Each stage has a specific responsibility.

Pattern Agent

Detects clinical/coding patterns, missing internal-review items, and documentation triggers.

Query Planner

Creates case-specific online retrieval queries for clinical, policy, coding, and medical necessity evidence.

Evidence Quality

Checks whether retrieved evidence is adequate and can trigger one retry when evidence is weak.

Verification Agent

Separates public evidence from internal checks such as payer policy, authorization, provider/NPI, and benefits.

Decision Intelligence

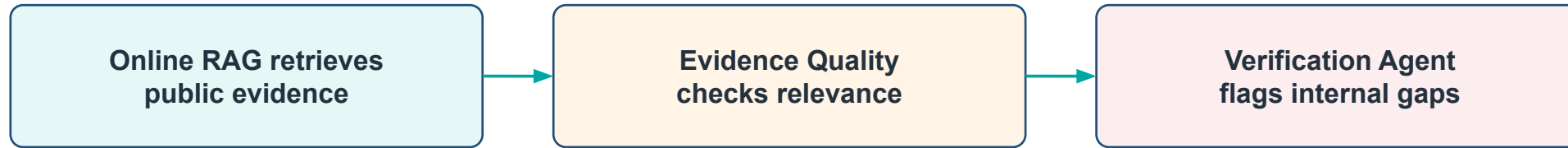
Synthesizes patterns + evidence + verification status into a reviewer-oriented recommendation.

Report Agent

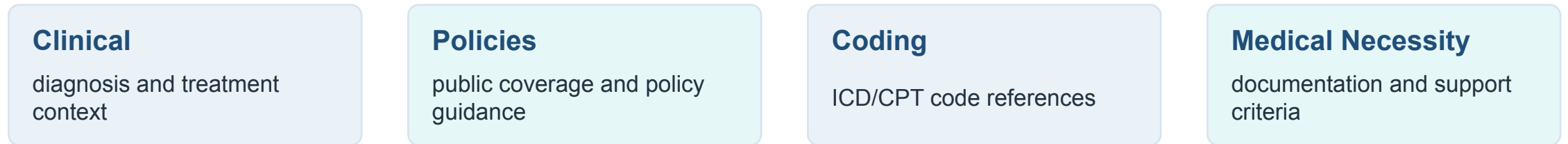
Creates a concise PDF report with evidence, verification gaps, and suggested reviewer actions.

RAG and Verification-Aware Design

Public evidence helps, but internal verification remains necessary.



Knowledge categories used in the POC



Key transparency principle: ClinIQ never treats missing internal data as if it were verified.

Prototype Demonstration Flow

Streamlit is used for a simple, understandable live demo.

1. Select case

Choose one of 10 structured claim JSON demo cases.

2. Run agents

The workflow shows progress through pattern, retrieval, verification, decision, and report stages.

3. Inspect output

Review evidence, verification status, recommended next action, and download the PDF.

Demo narrative

The demo should use CASE-HIP-006. It shows an orthopedic claim where public evidence supports the clinical relationship, but payer/provider/authorization checks require internal verification. The system recommends manual review, not approval.

What to show live

Workflow progress logs

Evidence retrieved and categorized

PDF report download

Generated Decision Intelligence Report

The final artifact is reviewer-oriented and transparent.

ClinIQ: Agentic Clinical Pattern Recognition and Decision Support Platform

Case ID: **CASE-HIP-006** | Generated: 2026-07-02 01:20
Evidence-analysis support only — human reviewer retains final decision authority.

Claim Key Details

Claim Detail	Value
Payer / Plan	National Health Plan / PPO
Policy ID	POL-STANDARD-2026
Provider / NPI	Metro General Hospital / 1234567890
Claim Type / POS	8371 / 21
Service Dates	2026-05-15 to 2026-05-18
Total Billed	\$28,500.00

Patterns Recognized

- 1 diagnosis code(s) linked to 2 procedure code(s).
- Diagnosis-to-procedure mapping is available and requires evidence review.
- Orthopedic procedure pattern detected: imaging, operative notes, and medical necessity documentation should be reviewed.
- 2 supporting document(s) available for review.
- Procedure documentation pattern detected: operative/procedure note is referenced.

Online Evidence Retrieved

Evidence Category	Online Source Used
Coding Evidence	ICD-10 M16.11 - Unilateral primary osteoarthritis, right hip ... (mirahealth.care)
Medical Necessity Evidence	ICD-10 M16.11 - Unilateral primary osteoarthritis, right hip Mira Health (mirahealth.care)
Coding Evidence	ICD-10 M16.11 - Unilateral primary osteoarthritis, right hip Mira Health (mirahealth.care)
Policy / Coverage Evidence	Article - Billing and Coding: Total Hip Arthroplasty (A57683) - CMS (cms.gov)
Clinical Evidence	CPT Code for Total Hip Arthroplasty 27130 Explained - MedHeave (medheave.com)
Clinical Evidence	I.D. Partial/Total Hip Replacements. Stride to Coding Success - AAPC (aapc.com)
Policy / Coverage Evidence	Total Hip Replacement Surgery CPT Code 27130 (orthobillingexpert.com)

Internal Verification Required

Clinical, coding, and general policy evidence can be evaluated from public sources, but payer, provider, or authorization checks require internal verification before final claim action.

Verification Check	Status / Reviewer Action
Payer Policy Verification	Internal verification required: Verify payer-specific policy, member benefits, exclusions, and coverage limits inside the payer portal or internal policy repository.
Provider / Hospital Verification	Internal verification required: Verify facility status, network status, NPI, billing provider role, rendering provider role, and specialty alignment internally.
Authorization Verification	Not publicly verifiable: Check internal authorization, admission, and member benefit systems.

Evidence Analysis Summary

The claim for total hip arthroplasty (CPT 27130) is supported by clinical evidence, including diagnosis of primary osteoarthritis of the right hip (M16.11) and medical necessity documentation. However, there are gaps in verification, including payer policy, provider status, and prior authorization, which require internal verification. The online evidence retrieved provides guidance on coding, medical necessity, and

policy coverage, but does not replace the need for internal review.

Clinical Evidence Assessment

Supportive Clinical Evidence Found — Pending Internal Verification

Explanation: The available clinical and coding evidence appears to support the diagnosis-to-procedure relationship and the documented treatment pattern. This only reflects clinical evidence support and does not represent claim approval, payment approval, or final adjudication.

Verification Status

Internal Verification Required

Explanation: Claim-specific payer policy, provider/NPI status, member benefits, and authorization records usually cannot be verified through public web evidence. These checks must be completed through internal payer, provider, or authorization systems before final claim action.

Key Evidence Supporting the Recommendation

- Clinical evidence supports the diagnosis of primary osteoarthritis of the right hip (M16.11) as documented in the orthopedic operative report and PT evaluation.
- Medical necessity is established through documentation of functional impact, such as gait disturbance and activity limitation, as required by payers for total hip arthroplasty.
- CPT code 27130 is correctly used for primary total hip arthroplasty, as confirmed by coding resources, including OrthoBillingExpert and MedHeave.
- Policy coverage evidence from CMS and other sources indicates that total hip arthroplasty is a covered service, but specific payer policy and member benefits must be verified.

Recommended Next Action

Requires Manual Review

Explanation: Continue internal payer, provider, and authorization verification. If internal verification confirms payer coverage, authorization status, and provider eligibility, the claim may proceed through the organization's standard adjudication workflow.

Suggested Reviewer Actions

- Verify payer-specific policy, member benefits, exclusions, and coverage limits inside the payer portal or internal policy repository.
- Check internal authorization, admission, and member benefit systems to confirm prior authorization status.
- Confirm facility status, network status, NPI, billing provider role, rendering provider role, and specialty alignment for Metro General Hospital.

Disclaimer: This report provides decision intelligence support only. It does not approve, deny, or adjudicate claims.

Report highlights

- Claim key details
- Patterns recognized
- Online evidence
- Internal verification
- Recommended next action

Transparency and Boundaries

The prototype is honest about what public RAG can and cannot verify.

Publicly retrievable

Clinical guidelines, coding references, public policy pages, medical necessity context.

Requires internal systems

Payer-specific benefits, authorization status, provider network status, member eligibility, contract terms.

Design choice

ClinIQ labels these gaps explicitly and recommends human verification before final action.

Responsible AI position

The project prioritizes transparency over unsupported automation. It is intentionally a decision-support platform, not a decision maker.

Future Roadmap

Natural extensions after the proof of concept.

Local vector KB

Curated payer policy and medical necessity repository

FHIR/EHR integration

Replace synthetic JSON with standardized healthcare data feeds

Historical analytics

Provider behavior, clustering, and time-series anomaly detection

Reviewer workbench

Case queue, audit trail, reviewer feedback, and final disposition capture

Important: these are future extensions, not claims about the current hackathon demo.

Conclusion

ClinIQ proves the concept with a practical, transparent workflow.

What the prototype demonstrates

Agentic workflow for healthcare decision support

Pattern recognition over structured claim data

Online RAG with evidence-quality checks

Verification-aware recommendations

PDF reporting for human reviewers

Thank you !!